COL1000: Lab 7

Semester 2, 2025–26

Instructions Upload your code to Gradescope as: `<entryNumber>_q<quesNumber>.py`. After uploading, run the autograder to check that all tests pass. Make sure not to include any custom prompt text in `input()` calls, and any extra text in `print()` statements. You must not use `math, ast` libraries and advanced built-in python functions such as `split`, `splitlines`, `map`.

Question 1: Sorting List Write a program that first reads a positive integer n, then reads n integers (one per line) and stores into a list, and finally outputs the list sorted in the non-decreasing order.

Example:

```
Input:
4
15
16
9
22

Output:
[9, 15, 16, 22]
```

Question 2: Sentence to Word List Write a program that reads a string (with no punctuation marks) and outputs a list containing all the words in the sentence. The words in the string are separated by single spaces.

Example:

```
Input:
Python programming is fun

Output:
['Python', 'programming', 'is', 'fun']
```

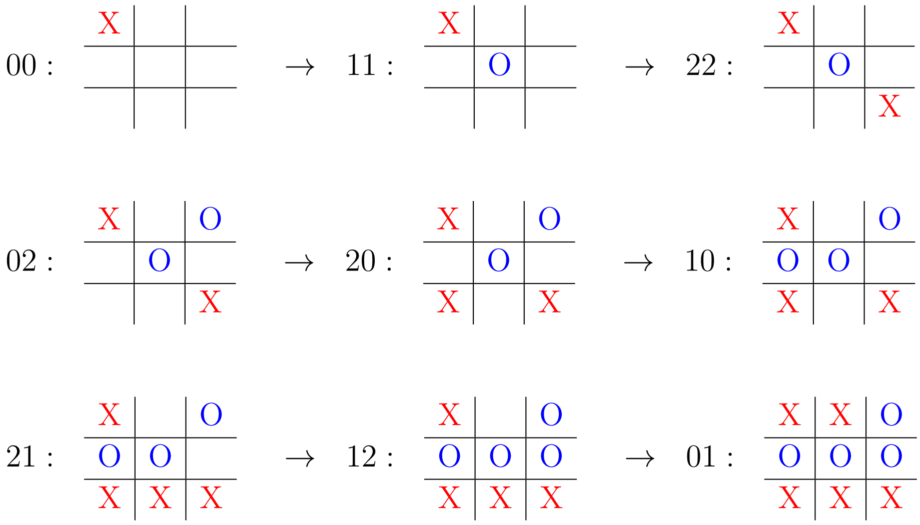
Question 3: Tic-Tac-Toe Simulate a 1-player Tic-Tac-Toe game in Python by reading 9 valid moves (one per line). Each move is a string of the form `"ab"`, where `a` and `b` represent the row and column of the move on a 3×3 board, using 0-based indexing (values $0, 1, 2$). Players alternately mark their move on the board, with player 1 moving first. After processing all moves, output:

- `0` if the game is a draw (i.e. no row, column, or diagonal fully occupied by a single player)

- 1 if Player 1 wins (i.e. player 1 is first to complete a full row, column, or diagonal)
- 2 if Player 2 wins (i.e. player 2 is first to complete a full row, column, or diagonal)

```
Input:
00
11
22
02
20
10
21
12
01

Output:
1
```



Question 4: Pascal's Triangle Write a program that reads a positive integer n, construct Pascal's Triangle up to row n, and output the elements of the last row as a list (assuming the 0^{th} row is the first row).

Example:

```
Input:
4

Output:
[1, 4, 6, 4, 1]
```

Explanation: Pascal's Triangle is a triangular arrangement of numbers where:

- The 0^{th} row has exactly one element - 1.
- The first and last element of every row is 1.
- The i^{th} element (other than the first / last element) of any row is equal to the sum of the i^{th} and $(i-1)^{th}$ element from the previous row.

The first 5 rows are:

```
    1
   1 1
  1 2 1
 1 3 3 1
1 4 6 4 1
```